

DataVinci Junior Analytics Engineer - What to Learn

Priority: MUST Learn (Class A keywords added to your resume)

- **BigQuery** — Google’s serverless data warehouse for running SQL over huge datasets. Your React/JS background means SQL is likely familiar territory; BigQuery is “SQL, but the engine scales for you.” Learn: standard SQL dialect quirks (backtick table refs like `project.dataset.table`), partitioned/clustered tables for cost control, and the BigQuery web console / `bq` CLI. Do the free Google Cloud Skills Boost “BigQuery Basics” lab (2-3 hrs).
- **Google Cloud Platform (GCP)** — the cloud ecosystem BigQuery lives in. You already know AWS (Secrets Manager) and some Azure concepts, so treat this as “a different vendor, same ideas” (IAM = permissions, Cloud Storage = S3, Cloud Functions = Lambda). Learn: GCP Console navigation, IAM basics, and how BigQuery/Cloud Storage/Compute Engine relate.
- **Power BI** — Microsoft’s dashboarding/BI tool, directly requested for building client-facing dashboards. If you’ve used Recharts/Mixpanel dashboards, the mental model (connect a data source → build visuals → publish) will feel familiar. Learn: Power Query (data shaping), DAX basics (calculated measures), and publishing/sharing a report. Microsoft’s free “Power BI Desktop” + their official “Get started” tutorial is the fastest path.
- **AWS CloudFormation** — infrastructure-as-code for AWS (you already use AWS Secrets Manager, so you know the console side). Learn: what a CloudFormation template (YAML/JSON) declares, stacks, and how it differs from clicking around the console. A basic understanding of “describe infra as code” is enough for a fresher role.
- **Database Testing** — validating data correctness (row counts, constraint checks, data-quality assertions) rather than testing application code. Learn: writing simple SQL-based validation queries (e.g., checking for nulls, duplicates, referential integrity) and the concept of a data-quality checklist. Your Skimly project’s hand-written schema work is a good bridge — you already think about schema correctness.
- **Data Engineering (as a discipline)** — the practice of building pipelines that move and transform data reliably. You’ve done pieces of this already (Skimly’s raw SQL pipeline, Clink’s Mixpanel event pipeline into dashboards) without the formal label. Learn: the ETL/ELT vocabulary (extract-transform-load), idempotency, and why schemas/data contracts matter.

Good to Know (Class B — best-effort, strengthens your fit)

- **Looker Studio** (formerly Google Data Studio) — free, web-based dashboard tool that connects directly to BigQuery/Sheets. Quick to pick up if you learn Power BI first; concepts transfer.
- **Microsoft Azure** — you have some exposure via Azure DevOps/App Services from other applications; the JD only needs general cloud-platform familiarity here.
- **API & JSON for data pipelines** — you already do this daily (REST APIs, Zod schemas); just reframe it as “fetching data for a pipeline” rather than “fetching data for a UI.”
- **Python for data formatting** — pandas basics (read.csv, groupby, merge) would be the highest-leverage 2-3 hours if you want to go beyond SQL-only data cleaning.

How this maps to your background

You are not starting from zero: PostgreSQL, SQL, REST APIs, and dashboard-building (Mixpanel → funnel/retention reports) are all things you’ve shipped in production or in a real project. The delta here is mostly **tooling** (BigQuery/GCP/Power BI are new UIs over concepts you already understand) rather than new fundamentals. Spend your prep time on BigQuery + Power BI first (highest keyword weight in the JD), then CloudFormation/database-testing vocabulary if time remains.